

V20PCA107-IT INFRASTRUCTURE MANAGEMENT

Unit-1: IT Infrastructure

Introduction, Challenges in IT Infrastructure Management, Design Factors for IT, Organizations, IT Infrastructures, IT Systems, Service Management Process Information systems Design Process, IT Infrastructure Library, Lab 1: Case Study and Hands-on training.

Week 1

Learning Objectives:

1. To understand the various components of IT Infrastructure
2. To comprehend the major Activities involved in IT infrastructure management
3. To identify the various challenges of managing IT infrastructure
4. To understand the factors to be considered for the Good Organizational Design
5. To understand about the necessity of Infrastructure design document

Key Topics

- 1.1 Definition of IT infrastructure
- 1.2 Need for the well planned IT infrastructure Management
 - 1.2.1 Major Aspects in IT infrastructure
 - 1.2.2 Aim of IT Infrastructure Management
 - 1.2.3 How to perform IT infrastructure Management
 - 1.2.4 Majors activities in IT infrastructure Management
- 1.3 “IT” For Business
- 1.4 Why it is challenging to manage IT infrastructure ?
- 1.5 Challenges in managing IT infrastructure
 - 1.5.1 Design issues of it organizations and it infrastructure
 - 1.5.2 Factors to be considered for the Good Organizational Design:

- 1.5.3 Design of IT Organization
 - 1.5.4 Success of any IT organization depends on its Infrastructure design
 - 1.6 Design of IT Organization
 - 1.6.1 Design of IT Organization (4 Assessments):
 - 1.7 Infrastructure design document
 - 1.7.1 The contents of the Design Document:
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1.1 Definition of IT infrastructure

IT infrastructure of an organization comprises of Hardware Equipments, Software, and any Other Components required delivering **IT services** to its customers.

1.2 Need for the well planned IT infrastructure Management

1. Almost all business activities of an organization highly depend on the infrastructure.
2. A good and reliable IT infrastructure is the key to the successful operations and is the foundation of any viable IT organization.
3. Every organization has to scale its information technology (IT) infrastructure to support business growth, managing global networks, databases and applications becomes an important task.
4. IT infrastructure needs to be developed and managed in an effective manner to support the requirements of the organization.

1.2.1 Major Aspects in IT infrastructure

1. Purchased software (e.g., ERP packages, RDBMS, operating systems, e-mail tools, office tools, financial applications, etc.)
2. IT infrastructure hardware (e.g., machines, desktops, servers, switches, communication devices, etc.)

3. Softwares developed
4. Software maintenance (corrective, perfective, adaptive, preventive)
5. IT services (e.g., software setup, help desk, computer administration, etc.)
6. Human resource (working staff)

1.2.2 Aim of IT Infrastructure Management

IT infrastructure management aims to manage these components for effective utilization in order to provide better services to customers.

Who will perform IT infrastructure Management

This is usually done by IT department of an organization in consultation with “top management”.

1.2.3 How to perform IT infrastructure Management

IT infrastructure management creates an infrastructure management environment that reduces IT complicacies.

It automates and supports required performance and service availability levels and resolves problems to ensure the business continuity.

1.2.4 Major Activities involved in IT infrastructure management

Among many other objectives, IT infrastructure management tries to:

1. Decrease the duplication of effort and increase organizational production
2. Decrease business risk
3. Ensure the use of standards
4. Ensure minimum downtime
5. Improve adaptability necessary for a changeable environment

6. Improve the information flow in information system
7. Ensure interoperability among organisational and external entities
8. Maintain effective change management policies and practices
9. Reduce operational costs
10. Increase service quality
11. Increase business planning

The following are the list of activities happening in the broad aspect of IT infrastructure management.

- IT infrastructure management aims to manage these components for effective utilization in order to provide better services to customers.
- Almost all business activities of an organization depend upon the infrastructure.
- Hence, its effective management is very essential.
- IT infrastructure needs to be developed and managed in an active manner to support the requirements of the organization.
- This is usually done by IT department of an organization in consultation with top management.
- A good and reliable IT infrastructure is the key to the successful operations and is the foundation of any viable IT organization.
- IT infrastructure management creates an infrastructure management environment that reduces IT complications.
- It automates and supports required performance and service availability levels and resolves problems to ensure the business continuity.

- As an organization scales its information technology (IT) infrastructure to support business growth, managing global networks, databases and applications becomes an important task.

1.3 “IT” For Business

IT department is playing a vital role for the development of any business organization

- IT Experts, Top management providing reliable services
- In the view of IT infrastructure the role of human IT expert is inevitable (Fig.1)
- The challenge of managing IT infrastructure is taken care by the IT expert
- Few important challenges are listed below:
 - suitability, low cost adaptability, Decrease business risk
- Hence the role of IT manager is really important in setting and managing any IT infrastructure
- *IT skills and Managerial practices* plays an important role in IT infrastructure management
- The management practices are developed over time and they are unique for each organization.

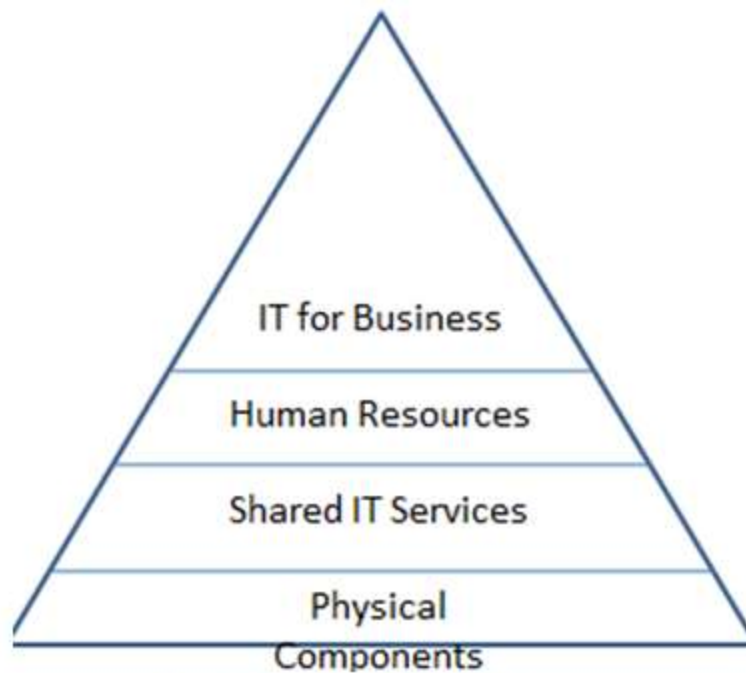


Fig.1 A View of IT infrastructure highlighting the importance of human element

- **shared services** : Ex: database services and connections to internet.
- Reliable IT services for business = Human Element + shared IT services + Physical components
- Good IT infrastructure management can avoid occurrence of IT infrastructure problems and resolve them before they impact business availability.
- IT skills and managerial practices also play an important role in IT infrastructure management
- The figure.1 (Ref fig.1) presents a view of IT infrastructure highlighting the importance of human element in it.
- Three bottom layers of this figure show the IT infrastructure components where bottom-most layer depicts physical shared components such as computers and common technologies.
- The second bottom layer contains the commonly shared services such as database services and connections to internet.

- The components of these two layers are changed into IT infrastructure services by human element using its skills, experience and knowledge.
- The human element binds IT components into a reliable set of IT infrastructure services which are shared in business processes across the organization.
- Good IT infrastructure management can avoid occurrence of IT infrastructure problems and resolve them before they impact business availability.
- The management aspect of IT infrastructure is very critical since management practices are developed over time and they are unique for each organization.
- IT infrastructure consists of resources and capabilities which are built through the interaction between technology and people in the organization.
- It consists of elements shared by different levels of users and processes and provides platform to the people to share knowledge.
- Though IT infrastructure and its management are very essential for smooth running of an organization, some challenges are faced in management and development of IT infrastructure.

1.4 Why it is challenging to manage IT infrastructure?

- IT infrastructure consists of different technology and people in the organization.
- It consists of elements shared by different levels of users and processes and provides platform to the people to share knowledge.

1.5 Few important challenges of managing IT infrastructure :

1. Suitability to the organization
2. Low cost and High Quality
3. Adaptability in Changeable environment
4. Decrease Business Risk

1. Suitability to the organization

- IT infrastructure management needs to develop and deploy management activities in such a way that they support operational and strategic goals of the organization.
- The management activity consists of two parts: Maintenance of existing systems and development of new infrastructure.
- Usually, information system development relies on existing hardware and software resources.
- However, vision is required in both business and IT in order to achieve what technology can do and how to make its best possible use.

2. Low cost and High Quality -

- Management activities should achieve low cost with High Quality.
- Sometimes, interactions and partnership with outside IT service providers may provide solutions to this challenge.

3. Adaptability in Changeable environment –

- IT infrastructure should not only be reliable in operations today but it should also be open for changes in the future to incorporate future business requirements.
- All choices that are made in developing the infrastructure are important because the infrastructure is an organizational asset and meant to be used for an extended period.

4. Decrease the Business Risk –

- Infrastructure provides the foundation upon which business applications are built.
- Therefore, it is required to be managed in such a way that it does not fail under any circumstances.
- Since these challenges are related to IT management problems, IT managers should develop management procedures to match the current and future requirements.

Why the Role of IT manager is really important?

Since the above challenges are related to IT management problems, IT managers should develop management procedures to match the current and future requirements.

1.5.1 Design issues of it organizations and it infrastructure

In the following section the major Design issues of it organizations and it infrastructure is discussed.

- The factors to be considered for the Good organizational Design
- Role of an IT Leader
- Designing IT organization (4 Assessments)

1.5.2 Factors to be considered for the Good Organizational Design

1. Business Requirements,
2. Necessary Infrastructure to support business operations

3. Good Strategy And Technology For The Necessary Deployment
4. Clearly Defined Accountability Plan

1.5.3 Design of IT Organization

1. no single proven optimal design strategy
2. IT leaders should seek the answer for four questions: what works, what does not work, when will it work and why?
3. but there are many best practices available for the learning and benchmarking purposes..
4. Keeping this challenge in mind, IT leaders always try to find out a perfect IT organizational model that addresses all problems in their own scenario.

Answers to these four questions provide the essence of a "best practice" for a specific organizational setting and help IT leaders in determining whether a specific "best practice" is appropriate for their organization or not.

1.5.4 Success of any IT organization depends on its Infrastructure design

1. With the business needs and the availability of effective and efficient IT infrastructure support.
2. To enable the business to function smoothly
3. Organizational design refers to the way in which an IT organization divides its work force into different tasks and operates by coordinating these tasks.
4. While carrying out design of an organization, major factors influencing organizational design must be carefully considered.
5. The organizational design must be measured to assess its effectiveness, strengths and weaknesses.

1.6 Design of IT Organization

The design of IT Organization is a challenging activity for any organization. Hence any proven design techniques need to be followed. But before designing such IT organization an exhaustive study about the organization's goal and existing structure must be analyzed.

1. Designing an effective organizational structure is a real challenge.
2. There Is No Single Optimal Organizational Design for IT Departments *

*[Courtesy : According to the people3, Inc., report Structuring for Success:
Building Blocks for IT Organization Design]

According to the Gartner's people3 organization, "There are four basic steps involved in the organizational design process to increase the rate of success of the IT re-engineering initiatives of any organization".

While designing the IT organization the following 4 assessment need to be done.

1. Business Driver Assessment
2. Organization Readiness Assessment
3. Organizational Structure Model Assessment
4. Business Impact Assessment

1.6.1 Design of IT Organization (4 Assessments):

1. Business Driver Assessment - This process identifies the business drivers that lead to the development of a re-engineering strategy.

2. Organization Readiness Assessment - It ensures that all the constraints and barriers to organizational re-engineering are evaluated and are taken into consideration during the design and implementation processes.

3. Structure Model Assessment - This process understands the strengths and weaknesses of each IT structure model (centralised, decentralised or hybrid).It

selects the organizational design that is not only aligned with business strategy but also fits to the culture of the organization.

4. Business Impact Assessment - This process conducts a series of "what if" business impact analyses during the organizational design phase. This helps in minimizing any potentially negative impact on the business and evaluates how well the new structure achieves the business and IT objectives.

1.7 INFRASTRUCTURE DESIGN DOCUMENT

In this section the following 4 aspects will be discussed.

1. Design Document
2. Content of Design document
3. IT infrastructure design should be able to take care of various tasks
4. Necessary to keep the business of an organization running smoothly.

Design of IT Infrastructure involves the designing the IT design document

- ✓ **Design document** = complete information about the IT infrastructure
- ✓ **Design Document** = written documentation of the design factors and the choices to satisfy the business and technical requirements.

The design documentation also aids in the implementation of the design. In many cases where the design architect is not responsible for the implementation, the design documents ensure the successful implementation of the design by the implementation engineer.

1.7.1 The contents of the Design Document:

1. Architecture design
2. Implementation plan
3. Installation guide
4. Validation test plan
5. Operational procedures

It documents the criteria that must be met to determine the success of the implementation and the test procedures that should be followed when validating the environment.

Design document contains following major aspects.

1. Design of Data Centre and Server Room
2. Design of IT network
3. Hardware and software specifications (Servers, Desktops and Laptops)
4. Specifications of Server and Client Operating systems
5. Details of access, controls to be implemented to access critical IT assets
6. Internet bandwidth, security devices and applications
7. Email service to be setup
8. Design of backup and disaster recover mechanism.

IT infrastructure design should be able to take care of various tasks necessary to keep the business of an organization running smoothly.

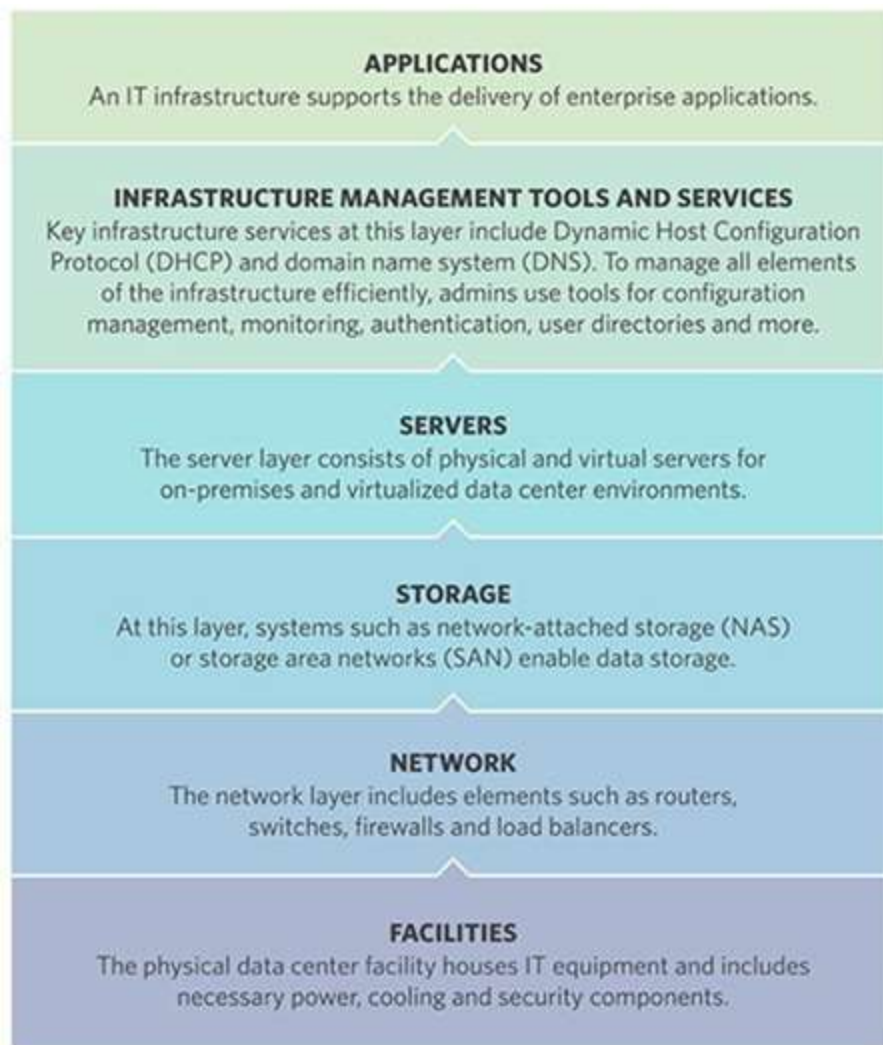


Fig.2 Layers of IT infrastructure

Summary

In this section the definition and various components of IT Infrastructure is given. The major activities happening under the umbrella term of IT infrastructure management is discussed. The various challenges of managing IT infrastructure and the factors to be considered for the Good Organizational Design are listed. The necessity of Infrastructure design document is also discussed.

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